

MTH203: Sequences and series

Tutorial/Homework 05

28/10/2025

Due date: 04/11/2025, 6:00 PM

Problem 1. Let $(a_n)_{n \in \mathbb{N}}$ be a sequence such that the sequence $(n^2 a_n)_{n \in \mathbb{N}}$ converges. Prove that the series $\sum_{n=1}^{\infty} a_n$ is convergent.

Problem 2. Determine whether the following series are convergent or divergent.

(a)

$$1 - \frac{1}{2^2} + \frac{1}{3} - \frac{1}{4^2} + \frac{1}{5} - \frac{1}{6^2} + \cdots$$

(b)

$$\frac{2}{2} - \frac{3}{4} + \frac{4}{6} - \frac{5}{8} + \frac{6}{10} - \frac{7}{12} + \cdots$$

Problem 3. Prove or disprove:

- (a) If $\sum_{n=1}^{\infty} a_n$ converges absolutely, then $\sum_{n=1}^{\infty} a_n^2$ also converges absolutely.
- (b) If the series $\sum_{n=1}^{\infty} a_n$ converges, and the sequence $(b_n)_{n \in \mathbb{N}}$ converges, then $\sum_{n=1}^{\infty} a_n b_n$ converges.